ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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INORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

s-Block Element

English Medium

EXERCISE # O-1

SELECT THE CORRECT ALTERNATIVE (ONLY ONE CORRECT ANSWER)

Alkali metals

1. Cs^+ ions impart violet colour to Bunsen flame. This is due to the fact that the emitted radiations are of -

(A) high energy (B) lower frequencies

(C) longer wave-lengths (D) zero wave number

SB0001

2. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then, A, B, C and D may be identified as

(A) Na, H_2, NaOH and Zn (B) K, H_2, KOH and Zn

(C) K, H_2, NaOH and Zn (D) Ca, H_2, $CaCOH_2$ and Zn

SB0002

3. Which of the following carbonate of alkali metals has the least thermal stability?

(A) Li_2CO_3 (B) K_2CO_3 (C) Cs_2CO_3 (D) Na_2CO_3

SB0003

4. The alkali metals which form normal oxide, peroxide as well as super oxides are

(A) Na, Li (B) K, Li (C) Li, Cs (D) K, Rb

SB0004

5. The pair of compounds, which cannot exist together in a solution is

(A) $NaHCO_3$ and NaOH (B) Na_2CO_3 and NaOH

(C) $NaHCO_3$ and Na_2CO_3 (D) $NaHCO_3$ and H_2O

SB0005

6. Solution of sodium metal in liquid ammonia is a strong reducing agent due to presence of

(A) solvated sodium ions (B) solvated hydrogen ions

(C) sodium atoms or sodium hydroxide (D) solvated electrons

SB0006

7. The order of solubility of lithium halides in non-polar solvents follows the order

(A) LiI > LiBr > LiCl > LiF (B) LiF > LiI > LiBr > LiCl

(C) LiCl > LiF > LiI > LiBr (D) LiBr > LiCl > LiF > LiI

SB0007

8. The salt which finds uses in qualitative inorganic analysis is

(A) $CuSO_4 \cdot 5H_2O$ or $ZnSO_4 \cdot 5H_2O$ (B) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$

(C) $Na(NH_4)HPO_4 \cdot 4H_2O$ (D) $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$

SB0008

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9. Fire extinguishers contain
(A) conc. H_2SO_4 solution (B) H_2SO_4 and $NaHCO_3$ solutions
(C) $NaHCO_3$ solution (D) $CaCO_3$ solution

SB0009

10. $CsBr_3$ contains
(A) Cs–Br covalent bonds (B) Cs^{3+} and Br^- ions
(C) Cs^+ and Br_3^- ions (D) Cs^{3+} and Br_3^{3-} ions

SB0010

11. $Na + Al_2O_3 \xrightarrow{\text{High temperature}} X \xrightarrow[\text{water}]{CO_2 \text{ in}} Y$; compound Y is
(A) $NaAlO_2$ (B) $NaHCO_3$ (C) Na_2CO_3 (D) Na_2O_2

SB0011

12. aq. $NaOH + P_4$ (white) $\longrightarrow PH_3 + X$; compound X is
(A) NaH_2PO_2 (B) $NaHPO_4$ (C) Na_2CO_3 (D) $NaHCO_3$

SB0012

13. When K_2O is added to water, the solution becomes basic in nature because it contains a significant concentration of
(A) K^+ (B) O^{2-} (C) OH^- (D) O_2^{2-}

SB0013

14. The order of melting point of chlorides of alkali metals is
(A) LiCl > NaCl > KCl < CsCl (B) LiCl > NaCl > KCl > CsCl
(C) NaCl > KCl > CsCl > LiCl (D) LiCl > NaCl > CsCl > KCl

SB0014

15. $NaOH(Solid) + CO \xrightarrow{200°C} X$; product X is
(A) $NaHCO_3$ (B) Na_2CO_3 (C) HCOONa (D) H_2CO_3

SB0015

16. The aqueous solutions of lithium salts are poor conductor of electricity rather than other alkali metals because of
(A) high ionisation energy
(B) high electronegativity
(C) lower ability of Li^+ ions to polarize water molecules
(D) higher degree of hydration of Li^+ ions

SB0016

17. In $LiAlH_4$, metal Al is present in
(A) anionic part (B) cationic part
(C) in both anionic and cationic part (D) neither in cationic nor in anionic part

SB0017

18. Which one of the following fluoride of alkali metals has the highest lattice energy?
(A) LiF (B) CsF (C) NaF (D) KF

SB0018

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19. Crown ethers and cryptands form
(A) complexes with alkali metals
(B) salts of alkali metals
(C) hydroxides of alkali metals used for inorganic quantitative analysis
(D) organic salts of alkali metals

SB0019

20. The correct order of degree of hydration of M^+ ions of alkali metals is
(A) $Li^+ < K^+ < Na^+ < Rb^+ < Cs^+$ (B) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
(C) $Cs^+ < Rb^+ < K^+ < Na^+ < Li^+$ (D) $Cs^+ < Rb^+ < Na^+ < K^+ < Li^+$

SB0020

21. The commercial method of preparation of potassium by reduction of molten KCl with metallic sodium at 850°C is based on the fact that
(A) potassium is solid and sodium distils off at 850 °C
(B) potassium being more volatile and distils off thus shifting the reaction forward
(C) sodium is less reactive than potassium at 850 °C with respect to Cl_2
(D) sodium has less affinity to chloride ions in the presence of potassium ion

SB0021

Alkaline earth metals

22. The 'milk of magnesia' used as an antacid is chemically
(A) $Mg(OH)_2$ (B) MgO (C) $MgCl_2$ (D) $MgO + MgCl_2$

SB0022

23. An alkaline earth metal (M) gives a salt with chlorine, which is soluble in water at room temperature. It also forms an insoluble sulphate whose mixture with a sulphide of a transition metal is called 'lithopone' -a white pigment. Metal M is
(A) Ca (B) Mg (C) Ba (D) Sr

SB0023

24. The hydroxide of IInd A metal, which has the lowest value of solubility product (K_{sp}) at normal temperature (25°C) is
(A) $Ca(OH)_2$ (B) $Mg(OH)_2$ (C) $Sr(OH)_2$ (D) $Be(OH)_2$

SB0024

25. Which of the following metal is inert towards reaction with H_2O.
(A) Be (B) Na (C) Ca (D) K

SB0025

26. $Mg_2C_3 + H_2O \longrightarrow$ X (organic compound). Compound X is
(A) C_2H_2 (B) CH_4 (C) propyne (D) ethene

SB0026

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27. The hydration energy of Mg^{2+} is

(A) more than that of Mg^{3+} ion (B) more than that of Na^{+} ion

(C) more than that of Al^{3+} ion (D) more than that of Be^{2+} ion

SB0027

28. The correct order of second ionisation potentials (IP) of Ca, Ba and K is

(A) K > Ca > Ba (B) Ba > Ca > K (C) K > Ba > Ca (D) K = Ba = Ca

SB0028

29. EDTA is used in the estimation of

(A) Mg^{2+} ions (B) Ca^{2+} ions

(C) both Ca^{2+} and Mg^{2+} ions (D) Mg^{2+} ions but not Ca^{2+} ions

SB0029

30. The correct order of solubility is

(A) $CaCO_3 < KHCO_3 < NaHCO_3$ (B) $KHCO_3 < CaCO_3 < NaHCO_3$

(C) $NaHCO_3 < CaCO_3 < KHCO_3$ (D) $CaCO_3 < NaHCO_3 < KHCO_3$

SB0030

31. The complex formation tendency of alkaline earth metals decreases down the group because

(A) atomic size increases

(B) availability of empty d and f-orbitals increases

(C) nuclear charge to volume ratio increases

(D) all the above

SB0031

32. The alkaline earth metals, which do not impart any colour to Bunsen flame are

(A) Be and Mg (B) Mg and Ca (C) Be and Ca (D) Be and Ba

SB0032

33. $Y \xleftarrow{\Delta, 205°C} CaSO_4 \cdot 2H_2O \xrightarrow{\Delta, 120°C} X$. X and Y are respectively

(A) plaster of paris, dead burnt plaster

(B) dead burnt plaster, plaster of paris

(C) CaO and plaster of paris

(D) plaster of paris, mixture of gases

SB0033

34. A metal M readily forms water soluble sulphate, and water insoluble hydroxide $M(OH)_2$. Its oxide MO is amphoteric, hard and having high melting point. The alkaline earth metal M must be

(A) Mg (B) Be (C) Ca (D) Sr

SB0034

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35. The purpose of addition of gypsum in the cement is -

(A) To slow down the process of setting of the cement

(B) To fasten the process of setting of the cement

(C) Not to affect the process of setting of the cement by any means

(D) None of these

SB0035

36. (Milky Cloud) C $\xleftarrow{CO_2}$ A + Na_2CO_3 $\longrightarrow$ B + C

The chemical formulae of A and B are

(A) NaOH and $Ca(OH)_2$ (B) $Ca(OH)_2$ and NaOH

(C) NaOH and CaO (D) CaO and $Ca(OH)_2$

SB0036

37. The correct order of basic-strength of oxides of alkaline earth metals is

(A) BeO > MgO > CaO > SrO (B) SrO > CaO > MgO > BeO

(C) BeO > CaO > MgO > SrO (D) SrO > MgO > CaO > BeO

SB0037

38. $X \xrightarrow{N_2,\Delta} Y \xrightarrow{H_2O} Z(\text{colourless gas}) \xrightarrow[CuSO_4]{} T(\text{blue colour})$

Then, substances Y and T are

(A) Y = Mg_3N_2 and T = $CuSO_4 \cdot 5H_2O$ (B) Y = Mg_3N_2 and T = $CuSO_4 \cdot 4NH_3$

(C) Y = $Mg(NO_3)_2$ and T = CuO (D) Y = MgO and T = $CuSO_4 \cdot 4NH_3$

SB0038

39. Weakest base among KOH, NaOH, $Ca(OH)_2$ and $Zn(OH)_2$ is

(A) $Ca(OH)_2$ (B) KOH

(C) NaOH (D) $Zn(OH)_2$

SB0039

40. If X and Y are the second ionisation potentials of alkali and alkaline earth metals of same period, then -

(A) X > Y (B) X < Y (C) X = Y (D) X << Y

SB0040

41. In castner-kellner cell if cathode is made up of graphite instead of mercury, then product that will be obtained first at cathode will be -

(A) Na-amalgam (B) Na only (C) H_2 gas (D) NaOH

SB0041

42. Na^+ ion can form complex with which of the following ligand.

(A) Cryptands (B) Crown ether (C) Both (A) and (B) (D) None

SB0042

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43. $X + C + Cl_2 \xrightarrow[\text{of about } 1000\,K]{\text{High temperature}} Y + CO$; $Y + 2H_2O \rightarrow Z + 2HCl$

Compound Y is found in polymeric chain structure and is an electron deficient molecule. Y must be

(A) BeO (B) $BeCl_2$ (C) BeH_2 (D) $AlCl_3$

SB0043

44. $BeCl_2 + LiAlH_4 \longrightarrow X + LiCl + AlCl_3$

(A) X is LiH (B) X is BeH_2

(C) X is $BeCl_2 \cdot 2H_2O$ (D) None

SB0044

45. The order of thermal stability of carbonates of IIA group is

(A) $BaCO_3 > SrCO_3 > CaCO_3 > MgCO_3$

(B) $MgCO_3 > CaCO_3 > SrCO_3 > BaCO_3$

(C) $CaCO_3 > SrCO_3 > BaCO_3 > MgCO_3$

(D) $MgCO_3 = CaCO_3 > SrCO_3 = BaCO_3$

SB0045

46. A pair of substances which gives all the same products on reaction with water is

(A) Mg and MgO (B) Sr and SrO (C) Ca and CaH_2 (D) Be and BeO

SB0046

47. A metal which is soluble in both water and liquid NH_3 separately -

(A) Cr (B) Mn (C) Ba (D) Al

SB0047

48. $Be_2C + H_2O \longrightarrow BeO + X$

$CaC_2 + H_2O \longrightarrow Ca(OH)_2 + Y$; then X and Y are respectively

(A) CH_4, CH_4 (B) CH_4, C_2H_6 (C) CH_4, C_2H_2 (D) C_2H_2, CH_4

SB0048

49. Which of the following groups of elements have chemical properties that are most similar

(A) Na, K, Ca (B) Mg, Sr, Ba (C) Be, Al, Ca (D) Be, Ra, Cs

SB0049

50. $MgBr_2$ and MgI_2 are soluble in acetone because of

(A) Their ionic nature (B) Their coordinate nature

(C) Their metallic nature (D) Their covalent nature

SB0050

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EXERCISE # O-2

SELECT THE CORRECT ALTERNATIVES (ONE OR MORE THAN ONE CORRECT ANSWERS)

Alkali metals

1. Nitrogen dioxide can be prepared by heating-

(A) KNO_3 (B) $AgNO_3$ (C) $Pb(NO_3)_2$ (D) $Cu(NO_3)_2$

SB0051

2. Which of the following compounds are not paramagnetic in nature?

(A) KO_2 (B) K_2O_2 (C) Na_2O_2 (D) RbO_2

SB0052

3. The golden yellow colour associated with NaCl to Bunsen flame can be explained on the basis of

(A) low ionisation potential of sodium

(B) emission spectrum

(C) photosensitivity of sodium

(D) sublimation of metallic sodium of yellow vapours

SB0053

4. KO_2 finds use in oxygen cylinders used for space and submarines. The fact(s) related to such use of KO_2 is/are

(A) it produces O_2 (B) it produces O_3

(C) it absorbs CO_2 (D) it absorbs both CO and CO_2

SB0054

5. The compound(s) which have –O–O– bond(s) is/are

(A) BaO_2 (B) Na_2O_2 (C) CrO_5 (D) Fe_2O_3

SB0055

6. Highly pure dilute solution of sodium in ammonia

(A) shows blue colouration due to solvated electrons

(B) shows electrical conductivity due to both solvated electrons as well as solvated sodium ions

(C) shows red colouration due to solvated electrons but a bad conductor of electricity

(D) produces hydrogen gas or carbonate

SB0056

7. Sodium metal is highly reactive and can be stored under

(A) toluene (B) kerosene oil (C) alcohol (D) benzene

SB0057

Alkaline earth metals

8. The compound(s) of IInd A metals, which are amphoteric in nature is/are

(A) BeO (B) MgO (C) $Be(OH)_2$ (D) $Mg(OH)_2$

SB0058

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9. The correct statement is/are

(A) $BeCl_2$ is a covalent compound

(B) $BeCl_2$ is an electron deficient molecule

(C) $BeCl_2$ can form dimer

(D) the hybrid state of Be in $BeCl_2$ is sp^2

SB0059

10. Which of the following substance(s) is/are used in laboratory for drying purposes?

(A) anhydrous P_2O_5 (B) graphite

(C) anhydrous $CaCl_2$ (D) Na_3PO_4

SB0060

11. Na_2SO_4 is water soluble but $BaSO_4$ is insoluble because

(A) the hydration energy of Na_2SO_4 is higher than that of its lattice energy

(B) the hydration energy of Na_2SO_4 is less than that of its lattice energy

(C) the hydration energy of $BaSO_4$ is less than that of its lattice energy

(D) the hydration energy of $BaSO_4$ is higher than that of its lattice energy

SB0061

12. Which of the following statements are false?

(A) $BeCl_2$ is a linear molecule in the vapour state but it is polymeric form in the solid state

(B) Calcium hydride is called hydrolith.

(C) Carbides of both Be and Ca react with water to form acetylene

(D) Oxides of both Be and Ca are amphoteric.

SB0062

13. Which of the following are ionic carbides?

(A) CaC_2 (B) Al_4C_3 (C) SiC (D) Be_2C

SB0063

14. Which of the following orders are **CORRECT** :

(A) AgCl > AgF : Covalent character order

(B) BaO > BaF_2 : Melting point order

(C) BeF_2 > BaF_2 : Solubility order

(D) $LiNO_3$ < $RbNO_3$: Thermal stability order

SB0064

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15. Which of the following statements are **CORRECT** :

(A) Mg is present in chlorophyll

(B) Alkaline earth metals does not form super oxide

(C) $NaHCO_3$ is known as baking soda

(D) Permanent hardness of water is removed by boiling

SB0065

16. Which of the following carbides on hydrolysis does not form methane :

(A) Be_2C (B) CaC_2 (C) SrC_2 (D) Mg_2C_3

SB0066

17. Select the incorrect order for given properties :

(A) Thermal stability : $BaSO_4 > SrSO_4 > CaSO_4$

(B) Solubility : $BaSO_4 > SrSO_4 > CaSO_4$

(C) Thermal stability : $Li_2CO_3 < Na_2CO_3 < K_2CO_3$

(D) Solubility : $Li_2CO_3 > Na_2CO_3 > K_2CO_3$

SB0067

18. The correct statement(s) is/are

(A) Mg cannot form complexes

(B) Be can form complexes due to a very small atomic size

(C) the first ionisation potential of Be is higher than that of Mg

(D) Mg forms an alkaline hydroxide while Be forms amphoteric oxides

SB0068

19. Which of the following is are the characteristic of barium?

(A) It produce water soluble sulphide, sulphite and sulphate

(B) It is a silvery white metal

(C) It forms $Ba(NO_3)_2$ which is used in preparation of green fire

(D) It produce blue-black solution in liquid ammonia

SB0069

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EXERCISE # S-1

NUMERIC GRID TYPE QUESTIONS :

1. Find the number of compounds from the following in which the element in the anionic part is in the minimum oxidation state of it

$LiH,\ Mg_3Bi_2,\ Al_4C_3,\ Ca_3P_2,\ BaO_2$

SB0070

2. How many nitrate groups are present in 1 molecule of Basic beryllium nitrate?

SB0071

3. Consider the following order :

(1) $CH_4 < CCl_4 < CF_4$: E.N. of central atom C

(2) $Mg^{+2} < K^+ < S^{-2} < Se^{-2}$: Ionic radius

(3) $Be^{+2}_{(aq)} > Mg^{+2}_{(aq)} > Ca^{+2}_{(aq)}$: Ionic mobility

(4) $Be^{+2} > Li^+ > Al^{+3}$: Hydrated size

(5) Be > Li > Cs : Reducing power

(6) $F^{\ominus}_{(aq)} > Cl^{\ominus}_{(aq)} > Br^{\ominus}_{(aq)}$: Electrical conductance in infinite dilute solution

Then calculate value of $|x - y|^2$, where x and y are correct and incorrect orders respectively.

SB0072

4. Consider the following elements :

Li, Cs, Mg, Pb, Al, N

- x = number of elements which can form MO type of oxides.
- y = the highest oxidation state shown by any one of them.
- z = the number of elements which can form amphoteric oxide(s).

Find the sum of x, y and z.

SB0073

Fill your answer as sum of digits till you get the single digit answer.

5. Find the number of s-block elements which can produce ammoniated cation and ammoniated electron with liquid ammonia.

Li, Na, K, Rb, Cs, Ca, Sr, Ba

SB0074

6. How many of the following metal chlorides impart characteristic colour to the oxidising flame.

$LiCl,\ NaCl,\ KCl,\ BeCl_2,\ MgCl_2,\ CaCl_2,\ SrCl_2,\ BaCl_2$

SB0075

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EXERCISE # S-2

COMPREHENSION BASED QUESTIONS

Comprehension # 1

$A \xrightarrow{\Delta} B\ (oxide) + CO_2$

$B + H_2O \longrightarrow C$

$C + CO_2 \longrightarrow A\ (milky)$

$C + NH_4Cl \xrightarrow{\Delta} D\ (gas)$

$D + H_2O + CO_2 \longrightarrow E$

$E + NaCl \longrightarrow F$

$F \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

1. A is :

(A) $Ca(HCO_3)_2$ (B) $CaCO_3$ (C) CaO (D) Na_2CO_3

SB0076

2. B and C are :

(A) CaO, $Ca(OH)_2$ (B) $Ca(OH)_2$, $CaCO_3$

(C) $CaCO_3$, $Ca(OH)_2$ (D) $Ca(OH)_2$, CaO

SB0076

3. D, E and F are :

(A) NH_3, NH_4Cl, NH_4HCO_3 (B) NH_3, NH_4HCO_3, $NaHCO_3$

(C) NH_4HCO_3, Na_2CO_3, $NaHCO_3$ (D) None

SB0076

Comprehension # 2

Alkali metals readily react with oxyacids forming corresponding salts like M_2CO_3, $MHCO_3$, MNO_3, M_2SO_4 etc. with evolution of hydrogen. They also dissolve in liquid NH_3 but without the evolution of hydrogen. The colour of its dilute solution is blue but when it is heated and concentrated then its colour becomes bronze.

4. Among the nitrate of alkali metals which one can be decomposed to its oxide easily?

(A) $NaNO_3$ (B) KNO_3 (C) $LiNO_3$ (D) $RbNO_3$

SB0077

5. Among the carbonates of alkali metals which one has highest stability?

(A) Cs_2CO_3 (B) Rb_2CO_3 (C) K_2CO_3 (D) Na_2CO_3

SB0078

6. Which of the following statement about the sulphate of alkali metal is correct?

(A) Except Li_2SO_4 all sulphate of other alkali metals are soluble in water

(B) All sulphates of alkali metals except lithium sulphate forms alum.

(C) The sulphates of alkali metals cannot be hydrolysed.

(D) All of these

SB0079

7. Which of the following statement about solution of alkali metals in liquid ammonia is correct?

(A) The solution have strong oxidizing properties.

(B) Both the dilute solution as well as concentrated solution are paramagnetic in nature

(C) Charge transfer is the responsible for the colour of the solution

(D) None of these

SB0080

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8. Which metal bicarbonates does not exist in solid state?

(i) $LiHCO_3$ (ii) $Ca(HCO_3)_2$ (iii) $Zn(HCO_3)_2$

(iv) $NaHCO_3$ (v) $AgHCO_3$

(A) (i), (ii), (iii), (v) (B) (i), (ii), (iii) (C) (i), (ii), (v) (D) (ii), (iii), (iv)

SB0081

MATCH THE COLUMN :

9.

Column-I	Column-II
(A) Hydrolith	(P) Contain Ca
(B) Nitrolim	(Q) Used as a fertilizer
(C) Dolomite	(R) Used to prepare H_2
(D) Pearl's ash	(S) Contain potassium

SB0082

10.

Column-I	Column-II
(A) Metal sulphate $\xrightarrow{\Delta}$ metal oxide + SO_2 + O_2	(P) Ba
(B) Metal cation + K_2CrO_4 $\longrightarrow$ yellow ppt	(Q) Sr
(C) Metal + NH_3 $\xrightarrow{\text{(liquid)}}$ blue solution	(R) Na
(D) MCl_2 + conc. H_2SO_4 $\longrightarrow$ white ppt.	(S) Mg

SB0083

MATCH THE CODE :

11.

List–I	List–II
(P) CaH_2	(1) Paramagnetic anion
(Q) K_2O_2	(2) Homodiatomic, diamagnetic anion
(R) KO_2	(3) Neutral aqueous solution
(S) NaCl	(4) Gives hydrogen on hydrolysis

Codes :

	P	Q	R	S
(A)	3	2	1	4
(B)	4	2	3	1
(C)	4	3	2	1
(D)	4	2	1	3

SB0084

12.

Column-I	Column-II
(P) Solvay process used for	(1) NaCl
(Q) Evolve $CO_2\uparrow$ on heating	(2) Na_2O_2
(R) aq. soln. is neutral towards litmus	(3) $NaHCO_3$
(S) Oxone	(4) Na_2CO_3

Codes :

	P	Q	R	S
(A)	3,4	3	1	2
(B)	4,1	1	3	2
(C)	2,3	4	1	3
(D)	2,4	1	3	4

SB0085

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ASSERATION & REASONING :

Questions given below consist of two statements each printed as Assertion (A) and Reason (R); while answering these questions you are required to choose any one of the following four responses:

(A) if both (A) and (R) are true and (R) is the correct explanation of (A)

(B) if both (A) and (R) are true but (R) is not correct explanation of (A)

(C) if (A) is true but (R) is false

(D) if (A) is false and (R) is true

13. **Assertion** : Beryllium does not impart any characteristic colour to the bunsen flame.

Reason : Due to its very high ionization energy, beryllium requires a large amount of energy for exciation of the electrons.

SB0086

14. **Assertion** : In fused state, calcium chloride cannot be used to dry alcohol or NH_3.

Reason : Anhy. $CaCl_2$ is not a good desiccant.

SB0087

15. **Assertion** : Diagonal relationship is shown between Be and Al.

Reason : Ionic potential of Be is almost the same as that of Al.

SB0088

16. **Assertion** : Beryllium halides dissolve in organic solvents.

Reason : Beryllium halides are ionic in character.

SB0089

17. **Assertion** : $BeCl_2$ fumes in moist air.

Reason : $BeCl_2$ reacts with moisture to form HCl gas.

SB0090

18. **Assertion** : Calcium carbide on hydrolysis gives methane.

Reason : Calcium carbide contains C_2^{2-} anion.

SB0091

19. **Assertion** : When CO_2 is passed through lime water, it first turns milky and then the solution becomes clear when the passage of CO_2 is continued.

Reason : The milkiness is due to the formation of insoluble $CaCO_3$ which then changes to soluble $Ca(HCO_3)_2$ when excess of CO_2 is present.

SB0092

20. **Assertion** : $MgCO_3$ is soluble in water when a current of CO_2 is passed.

Reason : The solubility of $MgCO_3$ is due to the formation of $Mg(HCO_3)_2$.

SB0093

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll Ana-Heating Effs,d-Block-Ron of Soll Ana-(OC)\Eng\03_s-Block\01_Th+Ex.p65

MATCHING LIST TYPE 1 × 3 Q. (THREE LIST TYPE Q.)

The following column 1, 2, 3 represent elements of s block and their different oxide formation abilities. Answer the questions that follow

Column-1 - Elements of s-Block

Column-2 - Product formed on reaction with excess oxygen

Column-3 - Characteristics of species form on reaction with excess oxygen

Column - 1 Elements	**Column - 2 Product formed on reaction with excess oxygen**	**Column - 3 Characteristics of species form on reaction with excess oxygen**
(I) Na	(A) Superoxide	(P) Paramagnetic
(II) Ba	(B) Peroxide	(Q) Diamagnetic
(III) K	(C) Monooxide	(R) Bond order = 1.5
(IV) Ca	(D) Dioxide	(S) Bond order = 1

21. Which of the following is an **INCORRECT** match.

(A) (I), (B), (QS) (B) (II), (B), (QS) (C) (II), (A), (PR) (D) (IV), (C), (Q)

SB0094

22. Which of the following matches will result in species having magnetic moment equal to that of Mn^{+6}

(A) I, (B), QS (B) IV, B, (QS) (C) IV, (A), (PR) (D) III, (A), (PR)

SB0094

23. On reaction with oxygen, which of the following combination is possible

(A) I, (A, C), (P, Q) (B) I, (B), (Q)

(C) II, (A, B), (P, Q, S) (D) IV, (B, C), Q

SB0094

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\0-Salt Anal-Heating Eff-s,d-Block-Ron of Salt Anal-(OC)\Eng\03_s-Block\01_TH+Ex.p65

EXERCISE # JEE-MAIN

1. The products obtained on heating $LiNO_3$ will be :- **[AIEEE-2011]**

(1) $LiNO_2 + O_2$ (2) $Li_2O + NO_2 + O_2$

(3) $Li_3N + O_2$ (4) $Li_2O + NO + O_2$

SB0103

2. What is the best description of the change that occurs when $Na_2O(s)$ is dissolved in water ? **[AIEEE-2011]**

(1) Oxidation number of sodium decreases

(2) Oxide ion accepts sharing in a pair of electrons

(3) Oxide ion donates a pair of electrons

(4) Oxidation number of oxygen increases

SB0104

3. Which of the following on thermal-decomposition yields a basic as well as an acidic oxide ? **[AIEEE-2012]**

(1) NH_4NO_3 (2) $NaNO_3$ (3) $KClO_3$ (4) $CaCO_3$

SB0105

4. Fire extinguishers contain H_2SO_4 and which one of the following :-**[JEE MAIN-2012, Online]**

(1) $CaCO_3$ (2) $NaHCO_3$ and Na_2CO_3

(3) Na_2CO_3 (4) $NaHCO_3$

SB0106

5. Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point ? **[JEE MAIN-2012, Online]**

(1) RbCl (2) LiCl (3) KCl (4) NaCl

SB0107

6. Which one of the following will react most vigorously with water ? **[JEE MAIN-2012, Online]**

(1) Li (2) K (3) Rb (4) Na

SB0108

7. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour, Y is :- **JEE MAIN-2012, Online]**

(1) NH_3 (2) MgO (3) Mg_3N_2 (4) $Mg(NO_3)_2$

SB0109

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Sol Anal-Heating Eff.s,d-Block-Ron of Salt Anal(OC)\Eng\03_s-Block\01_Th+Ex.p65

8. The correct statement for the molecule, CsI_3, is : **[JEE(Main)-2014]**

(1) it contains Cs^{3+} and I^- ions

(2) it contains Cs^+, I^- and lattice I_2 molecule

(3) it is a covalent molecule

(4) it contains Cs^+ and I_3^- ions

SB0110

9. Which of the following statements about Na_2O_2 is **not** correct ? **[JEE MAIN-2014, Online]**

(1) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium

(2) It is diamagnetic in nature

(3) It is the super oxide of sodium

(4) It is a derivative of H_2O_2

SB0111

10. Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are : **[JEE MAIN-2014, Online]**

(1) RbCl and $MgCl_2$

(2) LiCl and RbCl

(3) $MgCl_2$ and $BeCl_2$

(4) RbCl and $BeCl_2$

SB0112

11. The correct order of thermal stability of hydroxides is : **JEE(Main)Online-2015]**

(1) $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2$

(2) $Mg(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Ba(OH)_2$

(3) $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

(4) $Ba(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Mg(OH)_2$

SB0113

12. Which of the alkaline earth metal halides given below is essentially covalent in nature :-

(1) $SrCl_2$ (2) $CaCl_2$ (3) $BeCl_2$ (4) $MgCl_2$

JEE(Main)Online-2015]

SB0114

13. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy ? **[JEE(Main)-2015]**

(1) $BaSO_4$ (2) $SrSO_4$ (3) $CaSO_4$ (4) $BeSO_4$

SB0115

14. The commercial name for calcium oxide is : **[JEE(Main)-2016]**

(1) Quick lime (2) Milk of lime (3) Limestone (4) Slaked lime

SB0116

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\0-Salt Anal-Heating Eff-s,d-Block-Ron of Salt Anal-(OC)\Eng\03_s-Block\01_TH+Ex.p65

15. The correct order of the solubility of alkaline-earth metal sulphates in water is : **[JEE(Main)-2016]**

(1) Mg < Sr < Ca < Ba (2) Mg < Ca < Sr < Ba

(3) Mg > Ca > Sr > Ba (4) Mg > Sr > Ca > Ba

SB0117

16. The main oxides formed on combustion of Li, Na and K in excess of air are respectively :

(1) Li_2O, Na_2O_2 and KO_2 (2) Li_2O, Na_2O and KO_2 **[JEE(Main)-2016]**

(3) LiO_2, Na_2O_2 and K_2O (4) Li_2O_2, Na_2O_2 and KO_2

SB0118

17. Both lithium and magnesium display several similar properties due to the diagonal relationship ; however, the one which is incorrect is : **[JEE(Main)-2017]**

(1) Both form basic carbonates

(2) Both form soluble bicarbonates

(3) Both form nitrides

(4) Nitrates of both Li and Mg yield NO_2 and O_2 on heating

SB0119

18. In KO_2, the nature of oxygen species and the oxidation state of oxygen atom are, respectively **[JEE(Main)ONLINE-2018]**

(1) Superoxide an –1/2

(2) Oxide and –2

(3) Peroxide and –1/2

(4) Superoxide and –1

SB0121

19. In the following reactions products(A) and (B), respectively , are : **[JEE(Main)-2020]**

$NaOH + Cl_2 \rightarrow$ (A) + side products

(hot and conc.)

$Ca(OH)_2 + Cl_2 \rightarrow$ (B) + side products

(dry)

(1) $NaClO_3$and $Ca(OCl)_2$

(2) NaOCl and $Ca(ClO_3)_2$

(3) $NaClO_3$and $Ca(ClO_3)_2$

(4) NaOCl and $Ca(OCl)_2$

SB0130

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soli Ana-Heating Effs,d-Block-Ron of Soli Ana-(OC)\Eng\03_s-Block\01_Th+Ex.p65

20. Match the following compounds (Column-I) with their uses (Column-II) : **[JEE(Main)-2020]**

S.No.	Column – I	S.No.	Column – II
(I)	$Ca(OH)_2$	(A)	casts of statues
(II)	NaCl	(B)	white wash
(III)	$CaSO_4.\frac{1}{2}H_2O$	(C)	antacid
(IV)	$CaCO_3$	(D)	washing soda preparation

(1) (I)-(D), (II)-(A), (III)-(C), (IV)-(B)

(2) (I)-(B), (II)-(C), (III)-(D), (IV)-(A)

(3) (I)-(C), (II)-(D), (III)-(B), (IV)-(A)

(4) (I)-(B), (II)-(D), (III)-(A), (IV)-(C)

SB0131

21. Among the following, number of metal/s which can be used as electrodes in the photoelectric cell is _____ (Integer answer) **[JEE(Main)-2021]**

(A) Li (B) Na (C) Rb (D) Cs

SB0132

22. Match List-I with List-II. **[JEE(Main)-2021]**

	List-I		**List-II**
(a)	Sodium Carbonate	(i)	Deacon
(b)	Titanium	(ii)	Castner-Kellner
(c)	Chlorine	(iii)	Van-Arkel
(d)	Sodium hydroxide	(iv)	Solvay

Choose the correct answer from the options given below :

(1) (a)→(iv), (b)→(iii), (c)→(i), (d)→(ii) (2) (a)→(i), (b)→(iii), (c)→(iv), (d)→(ii)

(3) (a)→(iv), (b)→(i), (c)→(ii), (d)→(iii) (4) (a)→(iii), (b)→(ii), (c)→(i), (d)→(iv)

SB0133

23. Match List-I with List-II **[JEE(Main)-2021]**

List-I	**List-II**
(a) $Ca(OCl)_2$	(i) Antacid
(b) $CaSO_4.\frac{1}{2}H_2O$	(ii) Cement
(c) CaO	(iii) Bleach
(d) $CaCO_3$	(iv) Plaster of paris

Choose the most appropriate answer from the options given below :

(1) a-i, b-iv, c-iii, d-ii (2) a-iii, b-ii, c-iv, d-i

(3) a-iii, b-iv, c-ii, d-i (4) a-iii, b-ii, c-i, d-iv

SB0134

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\05-Salt Anal+Heating Eff+s,d-Block+Ion of Salt Anal+(OC)\Eng\03_s-Block\01_Th+Ex.p65

24. Match **List I** with **List II** : **[JEE(Main)-2021]**

List-I		**List-II**
Elements		**Properties**
(a) Li	(i)	Poor water solubility of I^- salt
(b) Na	(ii)	Most abundant element in cell fluid
(c) K	(iii)	Bicarbonate salt used in fire extinguisher
(d) Cs	(iv)	Carbonate salt decomposes easily on heating

Choose the correct answer from the options given below :

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(2) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(3) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

SB0135

25. The correct order of melting point is : **[JEE(Main)-2022]**

(1) Be > Mg > Ca > Sr
(2) Sr > Ca > Mg > Be
(3) Be > Ca > Mg > Sr
(4) Be > Ca > Sr > Mg

SB0146

26. Match the **List-I** with **List- II.** **[JEE(Main)-2022]**

List-I	**List-II**
(Metal)	**(Emitted light wavelength (nm))**
(A) Li	(I) 670.8
(B) Na	(II) 589.2
(C) Rb	(III) 780.0
(D) Cs	(IV) 455.5

Choose the most appropriate answer from the options given below:

(1) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
(2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
(3) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(4) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

SB0147

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s-Soll And Heating Eff.s,d-Block-Ron of Soll Ana-(OC)\Eng\03_s-Block\01_Th+Ex.p65

EXERCISE # JEE-ADVANCED

1. The species that do not contain peroxide linkage are - **[JEE 1992]**

(A) PbO_2 (B) H_2O_2 (C) SrO_2 (D) BaO_2

SB0122

2. Read the following statement and explanation and answer as per the options given below :

Statement-1 : The alkali metals can form ionic hydrides which contain the hydride ion H^-.

Statement-2 : The alkali metals have low electronegativity ; their hydrides conduct electricity when fused and liberate hydrogen at the anode. **[JEE 1994]**

(A) Both 1 and 2 are true and 2 is the correct explanation of 1.

(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.

(C) 1 is true but 2 is false.

(D) 1 is false but 2 is true.

SB0123

3. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. **[JEE 1996]**

K_2CO_3(I) $MgCO_3$(II) $CaCO_3$(III) $BeCO_3$(IV)

(A) I < II < III < IV (B) IV < II < III < I

(C) IV < II < I < III (D) II < IV < III < I

SB0124

4. Property of all the alkaline earth metals that increase with their atomic number is - **[JEE 1997]**

(A) ionisation energy (B) solubility of their hydroxides

(C) solubility of their sulphate (D) electronegativity

SB0125

5. Highly pure dilute solution of sodium in liquid ammonia - **[JEE 1998]**

(A) shows blue colour (B) exhibits electrical conductivity

(C) produces sodium amide (D) produces hydrogen gas

SB0126

6. The set representing the correct order of first ionization potential is - **[JEE 2001S]**

(A) K > Na > Li (B) Be > Mg > Ca (C) B > C > N (D) Ge > Si > C

SB0127

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Asseration and Reason

7. This questions contains statement-1 (asseration) and statement-2 (reason) and has 4 choices (a), (b), (c) and (d) out of which only one is correct.

Statement-1 : Alkali metals dissolve in liquid ammonia to give blue solutions. because.

Statement-1 : Alkali metals is liquid ammonia give solvated species of the type $[M(NH_3)_n]^+$ (M = alkali metals). **[JEE 2007]**

(A) Both 1 and 2 are true and 2 is the correct explanation of 1.

(B) Both 1 and 2 are true but 2 is not the correct explanation of 1.

(C) 1 is true but 2 is false.

(D) 1 is false but 2 is true.

SB0128

8. The compound(s) formed upon combustion of sodium metal in excess air is (are) **[JEE 2009]**

(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH

SB0129

9. List-I contains compounds and List-II contains reaction **[JEE Adv. 2022]**

List-I	**List-II**
(I) H_2O_2	(P) $Mg(HCO_3)_2 + Ca(OH)_2$
(II) $Mg(OH)_2$	(Q) $BaO_2 + H_2SO_4 \rightarrow$
(III) $BaCl_2$	(R) $Ca(OH)_2 + MgCl_2$
(IV) $CaCO_3$	(S) $BaO_2 + HCl \rightarrow$
	(T) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow$

Match each compound in List- I with its formation reaction(s) in List-II, and choose the correct option

(A) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ S; IV $\rightarrow$ R (B) I $\rightarrow$ T; II $\rightarrow$ P; III $\rightarrow$ Q; IV $\rightarrow$ R

(C) I $\rightarrow$ T; II $\rightarrow$ R; III $\rightarrow$ Q; IV $\rightarrow$ P (D) I $\rightarrow$ Q; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ P

SB0145

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	A	D	A	D	A	C	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	C	C	C	D	A	A	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	C	D	A	C	B	A	C	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	A	A	B	A	B	B	B	D	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	C	B	B	A	C	C	C	B	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BCD	BC	AB	AC	ABC	AB	ABD	AC	ABC	AC
Que.	11	12	13	14	15	16	17	18	19	
Ans.	AC	CD	ABD	ABCD	ABC	BCD	BD	BCD	BCD	

EXERCISE # S-1

Que.	1	2	3	4	5	6	
Ans.	4	6	4	10 (OMR-1)	8	6	

Except Be & Mg other s-block metals impart characteristic colour to oxidising flame.

EXERCISE # S-2

Que.	1	2	3	4	5	6	7	8		
Ans.	B	A	B	C	A	D	D	A		
Que.	9				10				11	12
Ans.	A→PR ; B→PQ ; C→P ; D→S				A→PQS ; B→PQ ; C→PQR ; D→PQ				D	A
Que.	13	14	15	16	17	18	19	20	21	22
Ans.	A	C	A	C	A	D	A	A	C	D
Que.	23									
Ans.	B									

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	4	4	3	3	4	3	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	1	3	1	1	1	1	4
Que.	21	22	23	24						
Ans.	1	1	3	1						

25. (4)

Sol.

	M.P
Be	1560 K
Mg	924 K
Ca	1124 K
Sr	1062 K

26. (1)

Sol. NCERT Table 10.1.5

Metal	Li	Na	K	Rb	Cs
Colour	Crimson red	Yellow	Violet	Red Violet	Blue
λ /nm	670.8	589.2	766.5	780.0	455.5

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A	A	B	B	A,B	B	B	A,B	D	

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